THE SUPPORT OF A FINITE TYPE QUASI-COHERENT MODULE ON AN ALGEBRAIC SPACE IS CLOSED AND COMPUTES BY STALKS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof handles the three assertions in three paragraphs of one proof environment; the geometric point argument in the third paragraph is the same device as in the second.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The preimage of the support under an etale morphism is the support of the pullback. Lemma lemma-support-covering, used at proof lines 3075-3077. Reduces closedness of the support to the scheme case.
- The support of a finite type quasi-coherent module on a scheme is closed. Morphisms, Lemma morphisms-lemma-support-finite-type, used at proof lines 3076-3077. Is the scheme input for assertion (1).
- Nakayama's lemma. Algebra, Lemma algebra-lemma-NAK, used at proof lines 3081-3083. Gives the second equivalence in assertion (2).
- Stalk of a pullback of a quasi-coherent sheaf. Properties of Spaces, Lemma spaces-properties-lemma-stalk-pullback-quasi-coherent, used at proof lines 3090-3097. Computes the stalk of f^*F at a geometric point as a tensor product.

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07TY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9953–9953. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 056J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6712–6712. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05VQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7559–7559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 05VR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7560–7560. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module. Then*

(1) *The support of $\mathcal{F}$ is closed.*
(2) *For a geometric point $\overline{x}$ lying over $x \in |X|$ we have*
$$x \in \mathit{Supp}(\mathcal{F}) \Leftrightarrow \mathcal{F}_{\overline{x}} \neq 0 \Leftrightarrow \mathcal{F}_{\overline{x}} \otimes_{\mathcal{O}_{X,\overline{x}}} \kappa(\overline{x}) \neq 0.$$
(3) *For any morphism of algebraic spaces $f : Y \to X$ the pullback $f^*\mathcal{F}$ is of finite type as well and we have $\mathit{Supp}(f^*\mathcal{F}) = f^{-1}(\mathit{Supp}(\mathcal{F}))$.*

Proof. Choose a scheme U and a surjective étale morphism $\varphi : U \to X$. By Lemma 07TY the inverse image of the support of $\mathcal{F}$ is the support of $\varphi^*\mathcal{F}$ which is closed by Morphisms, Lemma 056J. Thus (1) follows from the definition of the topology on $|X|$.

The first equivalence in (2) is the definition of support. The second equivalence follows from Nakayama's lemma, see Algebra, Lemma 00DV.

Let $f : Y \to X$ be as in (3). Note that $f^*\mathcal{F}$ is of finite type by Properties of Spaces, Section 05VR. For the final assertion, let $\overline{y}$ be a geometric point of Y mapping to the geometric point $\overline{x}$ on X. Recall that
$$(f^*\mathcal{F})_{\overline{y}} = \mathcal{F}_{\overline{x}} \otimes_{\mathcal{O}_{X,\overline{x}}} \mathcal{O}_{Y,\overline{y}},$$
see Properties of Spaces, Lemma 05VQ. Hence $(f^*\mathcal{F})_{\overline{y}} \otimes \kappa(\overline{y})$ is nonzero if and only if $\mathcal{F}_{\overline{x}} \otimes \kappa(\overline{x})$ is nonzero. By (2) this implies $x \in \mathrm{Supp}(\mathcal{F})$ if and only if $y \in \mathrm{Supp}(f^*\mathcal{F})$, which is the content of assertion (3). □